

AI-Driven Classification and Clustering of Dry Beans for Smart Agriculture Using Machine Learning

Focus Area: Application of Artificial Intelligence (AI) for Agriculture Food Processing

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Motivation

- ✓ In order to keep the economic value of Dry beans which have an important place in the agricultural economy, the efficiency of post-harvest industrial processes is very important.
- ✓ To provide this efficiency, new methods and technologies are needed for the separation and classification of dry beans.
- ✓ Different dry beans address different markets, which increases the need for the classification of beans.
- ✓ In this study, it is aimed to develop a classification model different from traditional separation methods, based on image processing and artificial intelligence.



Problem Statement & Objectives



PROBLEM STATEMENT

Accurate identification and grouping of dry bean types is essential for quality control and inventory management. Manual classification is time-consuming and inconsistent.

The problem is to automatically classify dry bean types and discover natural groupings using their physical attributes.



PROPOSED SOLUTION

We propose a machine learning based approach that combines:



CLASSIFICATION

To predict the class (type) of a dry bean using its features.



CLUSTERING

To group dry beans into natural clusters based on similarity in features.



PERFORMANCE ANALYSIS

To evaluate different models using performance metrics and identify the best performing model.

OBJECTIVES

1



To collect and preprocess dry bean dataset with relevant features.

2



To build and evaluate classification models to predict dry bean types.

3



To apply clustering algorithms to discover natural groupings in dry beans.

4



To analyze and compare performance metrics to identify the best model.

5



To provide insights that can help in quality assessment and decision making.

PERFORMANCE METRICS

CLASSIFICATION METRICS



Accuracy



Precision



Recall



F1-Score



ROC-AUC

CLUSTERING METRICS



Silhouette Score



Davies-Bouldin Index

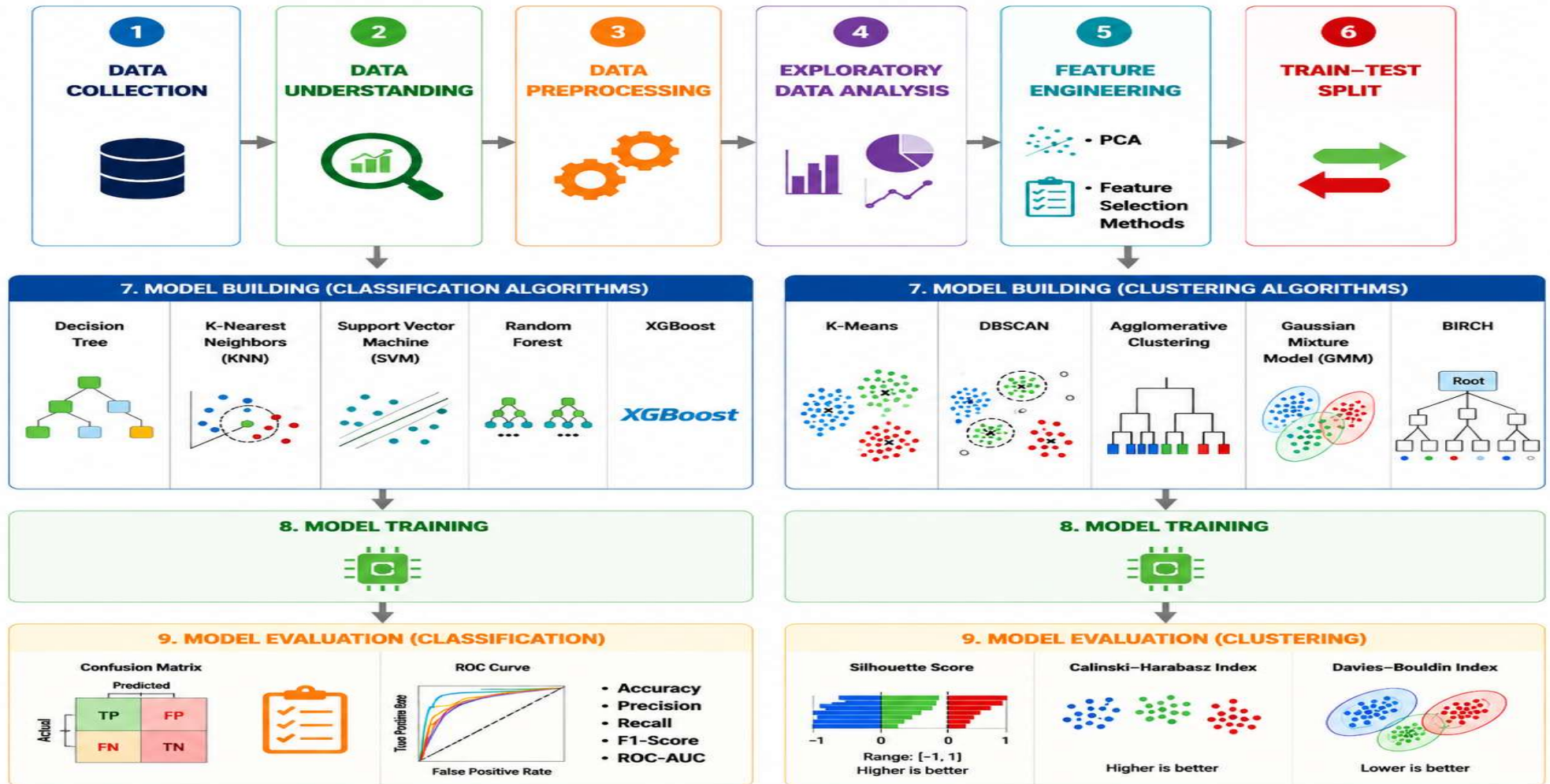


Calinski-Harabasz Score



By analyzing these metrics, we select the best classification model and the most meaningful clustering algorithm for effective dry bean analysis.

Methodology

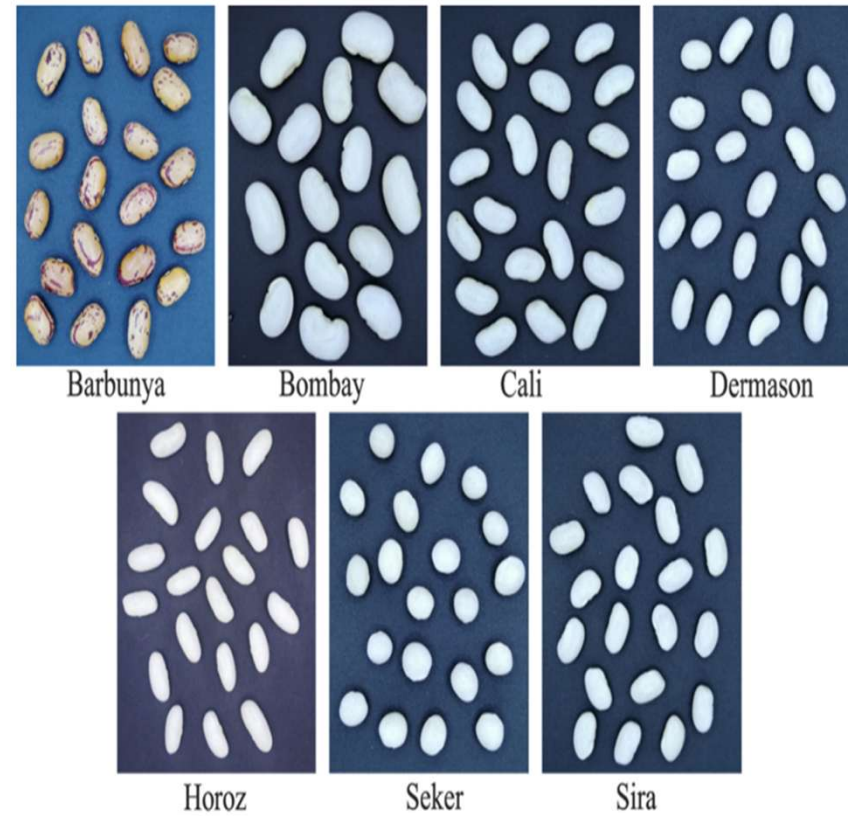


Classification

Clustering

Dataset Information

S. No.	Dry Bean Class	Quantity
1	Sira	2636
2	Horoz	1928
3	Call	1630
4	Dermason	3546
5	Bombay	522
6	Barbunya	1322
7	Seker	2027
	Total	13611



Feature Extraction



Morphological			Morphological / Shape		
S. No.	Feature	Expression	S. No.	Feature	Expression
M_1	Area (A)	πLW	M_9	Compactness (CO)	$\frac{E_d}{L}$
M_2	Equivalent diameter (E_d)	$\sqrt{\frac{4 * A}{\pi}}$	M_{10}	Minor axis length (l)	The length of the longest line $\perp$ to major axis (L)
M_3	Convex area (C)	The quantity of pixels inside the smallest convex polygon capable of containing the area of a single bean grain.	M_{11}	Aspect ratio (K)	$\frac{L}{l}$
M_4	Solidity (S)	$\frac{A}{C}$	M_{12}	Eccentricity (Ec)	$\sqrt{1 - \frac{l^2}{L^2}}$
M_5	Extent (E_x)	$\frac{A}{A_B}$ where A_B is area of bounding rectangle	S_1	Shape Factor 1	$\frac{L}{A}$
M_6	Perimeter (P)	Bean perimeter is measured as its border length	S_2	Shape Factor 2	$\frac{l}{A}$
M_7	Roundness (R)	$\frac{4\pi A}{P^2}$	S_3	Shape Factor 3	$\frac{4A}{\pi L^2}$
M_8	Major axis length (L)	The length of the longest line that can be traced from a bean.	S_4	Shape Factor 4	$\frac{4A}{\pi Ll}$

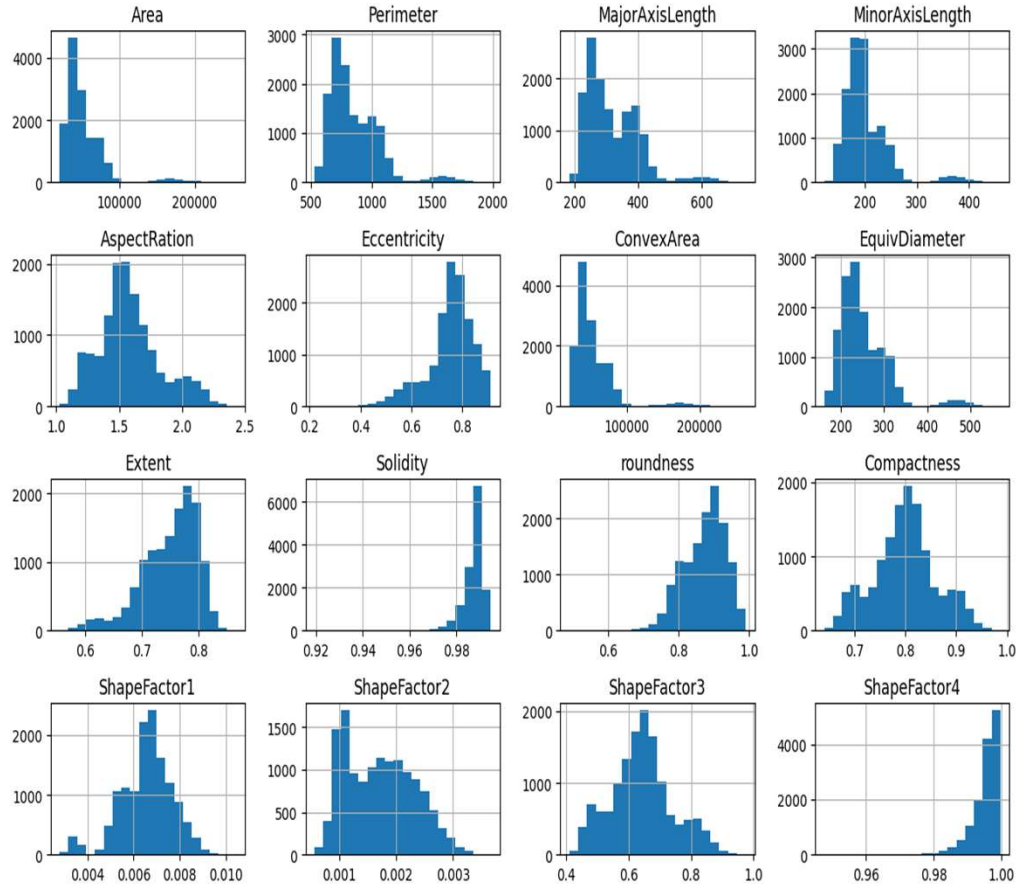
Feature Extraction



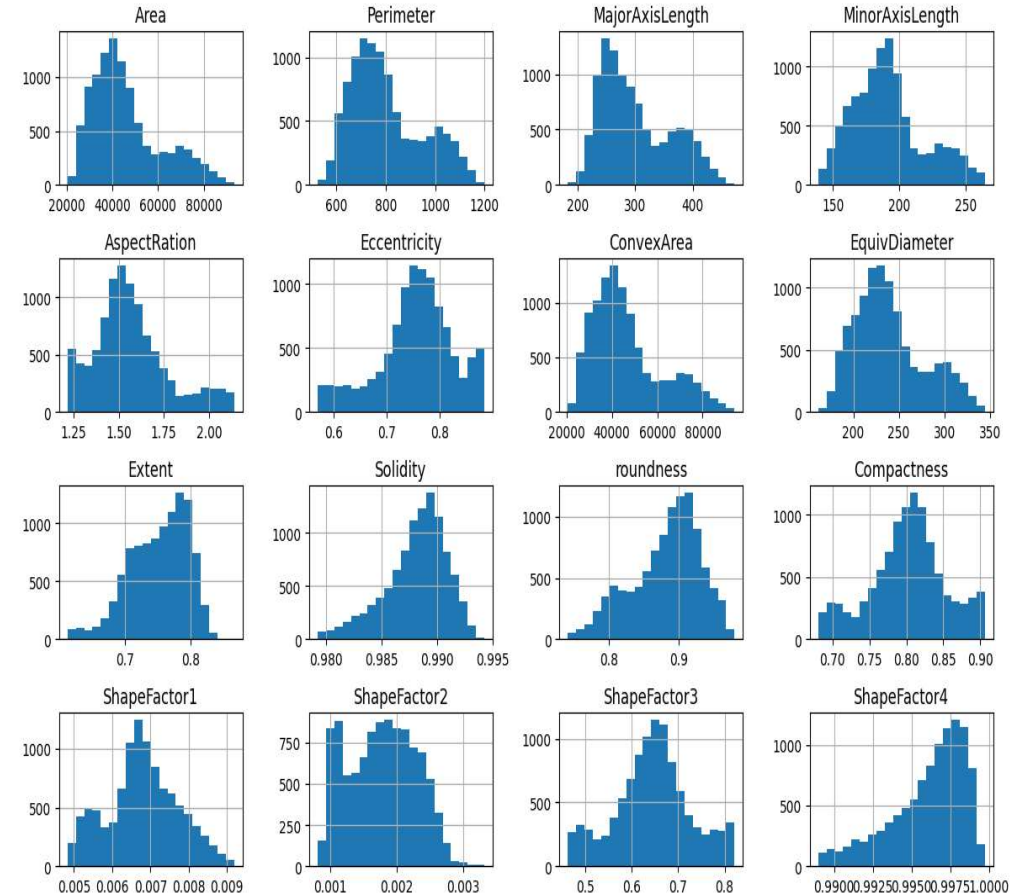
1	Area	Perimeter	MajorAxisLength	MinorAxisLength	AspectRatio	Eccentricity	ConvexArea	EquivDiameter	Extent	Solidity	roundness	Compactness	ShapeFactor1	ShapeFactor2	ShapeFactor3	ShapeFactor4	Class
2	28395	610.291	208.1781167	173.888747	1.197191424	0.549812187	28715	190.1410973	0.763922518	0.988855999	0.958027126	0.913357755	0.007331506	0.003147289	0.834222388	0.998723889	SEKER
3	28734	638.018	200.5247957	182.7344194	1.097356461	0.411785251	29172	191.2727505	0.783968133	0.984985603	0.887033637	0.953860842	0.006978659	0.003563624	0.909850506	0.998430331	SEKER
4	29380	624.11	212.8261299	175.9311426	1.209712656	0.562727317	29690	193.4109041	0.778113248	0.989558774	0.947849473	0.908774239	0.007243912	0.003047733	0.825870617	0.999066137	SEKER
5	30008	645.884	210.557999	182.5165157	1.153638059	0.498615976	30724	195.4670618	0.782681273	0.976695743	0.903936374	0.928328835	0.007016729	0.003214562	0.861794425	0.994198849	SEKER
6	30140	620.134	201.8478822	190.2792788	1.06079802	0.333679658	30417	195.896503	0.773098035	0.99089325	0.984877069	0.970515523	0.00669701	0.003664972	0.941900381	0.999166059	SEKER
7	30279	634.927	212.5605564	181.5101816	1.171066849	0.52040066	30600	196.3477022	0.775688485	0.989509804	0.943851783	0.923725952	0.007020065	0.003152779	0.853269634	0.999235781	SEKER
8	30477	670.033	211.0501553	184.0390501	1.146768336	0.489477894	30970	196.9886332	0.762401501	0.984081369	0.853079869	0.933373552	0.006924899	0.003242016	0.871186188	0.999048736	SEKER
9	30519	629.727	212.9967551	182.7372038	1.165590535	0.513759558	30847	197.1243203	0.770681818	0.989366875	0.967109244	0.925480392	0.006979152	0.003158285	0.856513956	0.99834456	SEKER
10	30685	635.681	213.5341452	183.1571463	1.165852108	0.51408086	31044	197.659696	0.771561479	0.988435769	0.954239808	0.925658498	0.00695891	0.00315155	0.856843654	0.998952981	SEKER
11	30834	631.934	217.2278128	180.8974686	1.2008339	0.553642225	31120	198.1390121	0.783682806	0.990809769	0.97027823	0.91212543	0.007045074	0.00300804	0.8319728	0.999061142	SEKER
12	30917	640.765	213.5600894	184.4398709	1.157884618	0.504102365	31280	198.4055115	0.770805285	0.988395141	0.94625818	0.929038343	0.006907529	0.00317422	0.863112242	0.999384389	SEKER
13	31091	638.558	210.4862549	188.3268476	1.117664622	0.446621924	31458	198.9630385	0.786377317	0.988333651	0.958172836	0.945254305	0.006770006	0.003333984	0.8935057	0.998639711	SEKER

EDA - Histograms

BEFORE CLEANING - Histograms

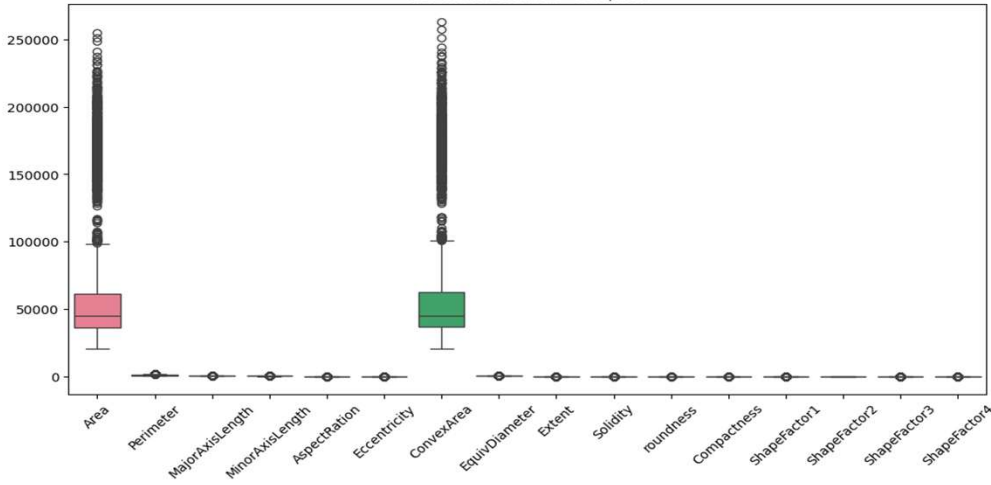


AFTER CLEANING - Histograms

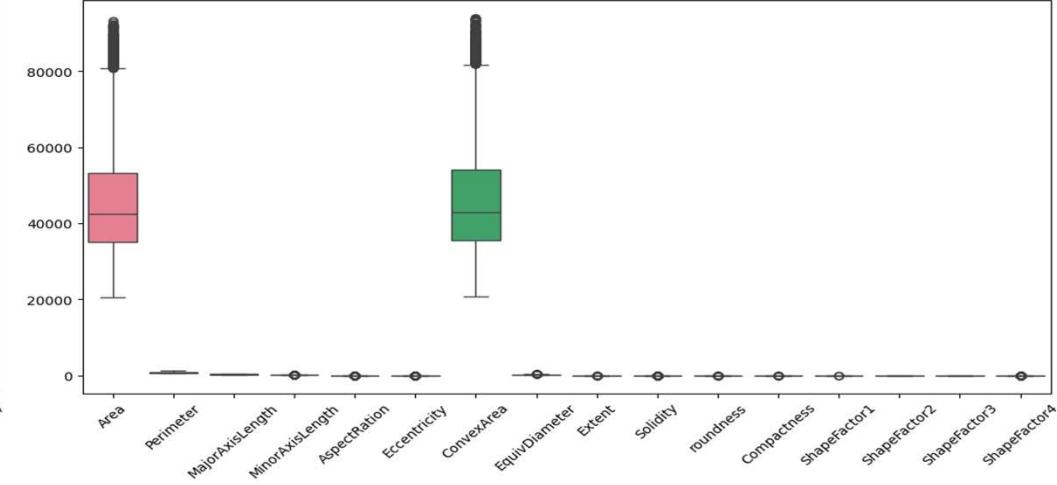


EDA- BoxPlots

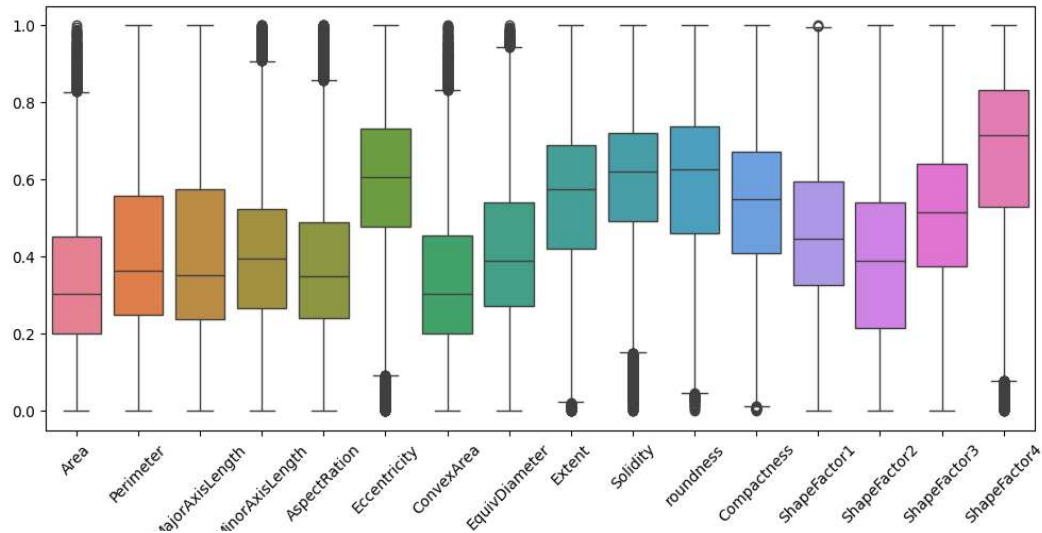
BEFORE CLEANING - Boxplots



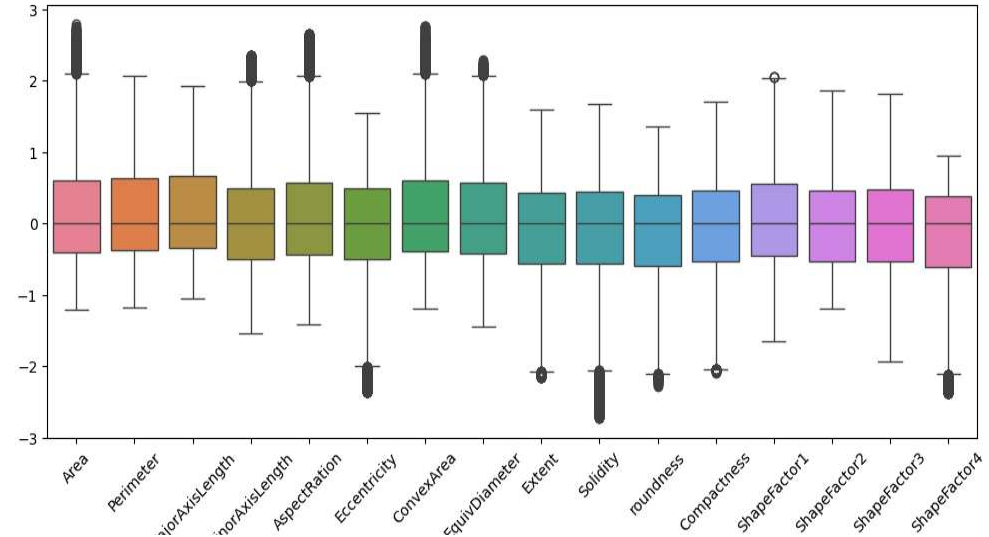
AFTER CLEANING - Boxplots



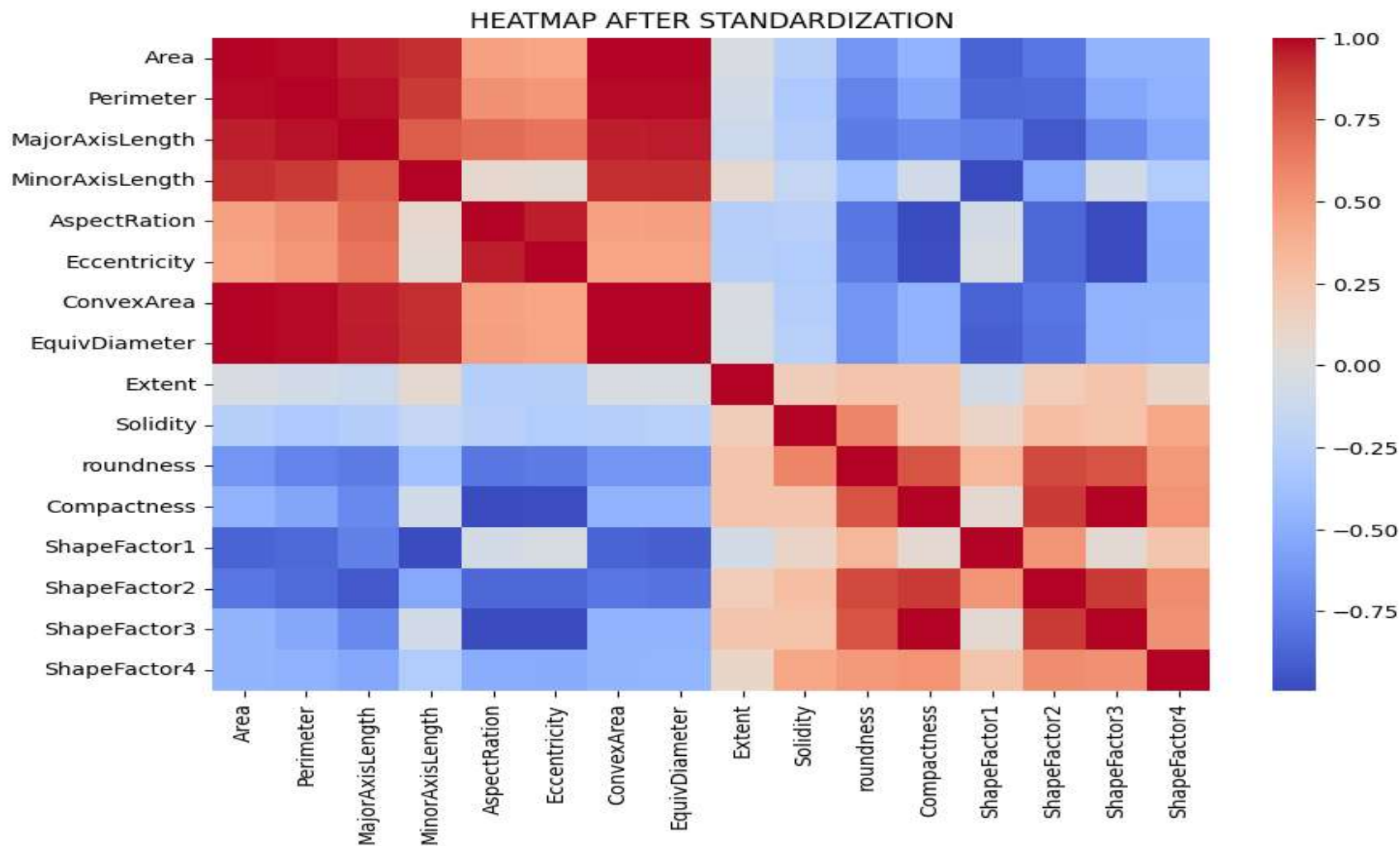
MIN-MAX SCALING



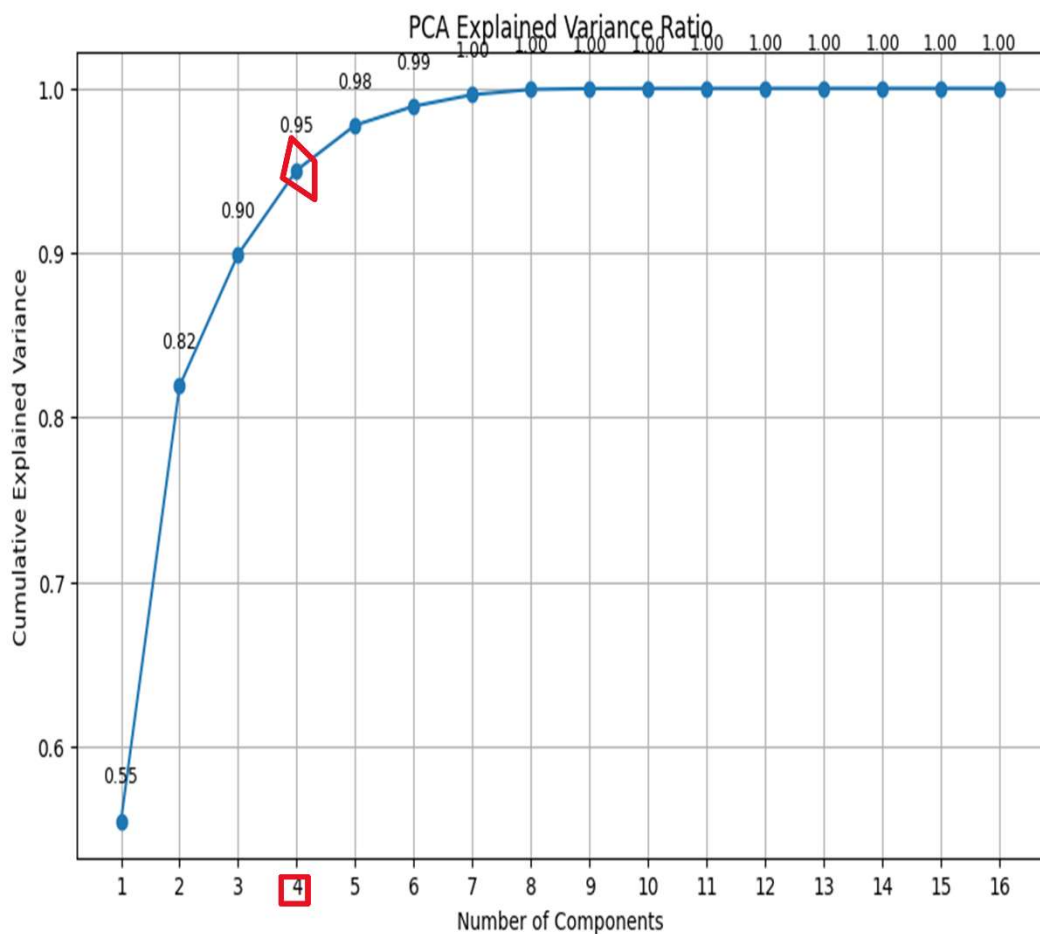
ROBUST SCALING



EDA - HeatMap



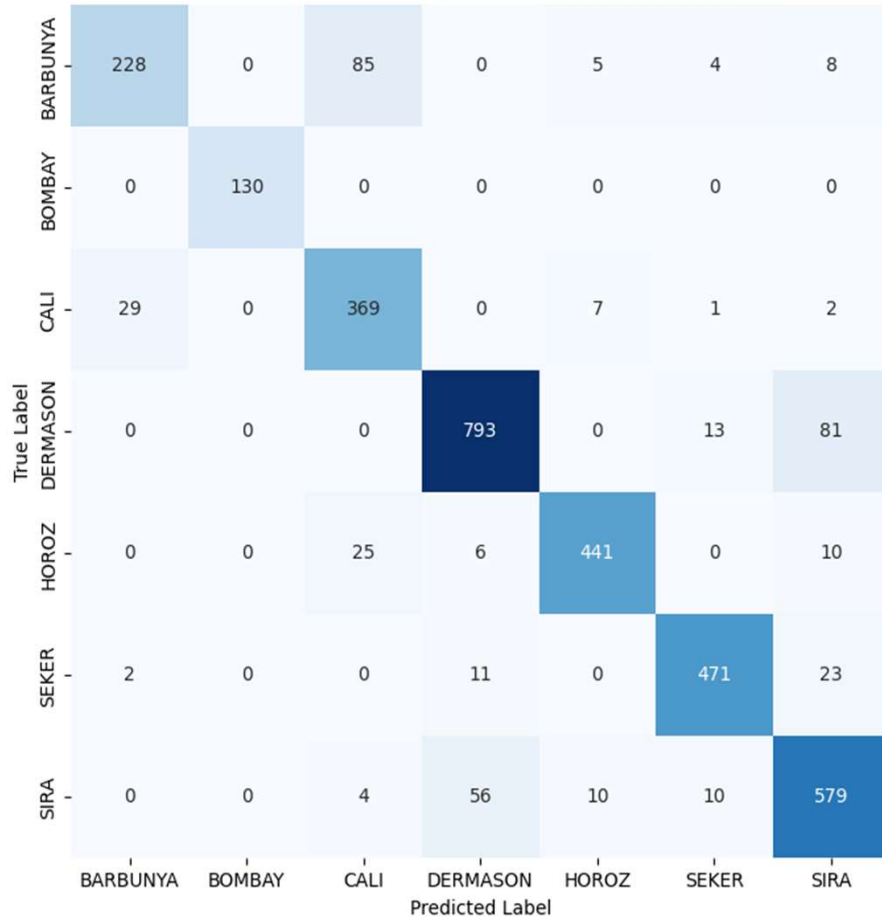
Feature Engineering – PCA, Feature Selection Algorithms



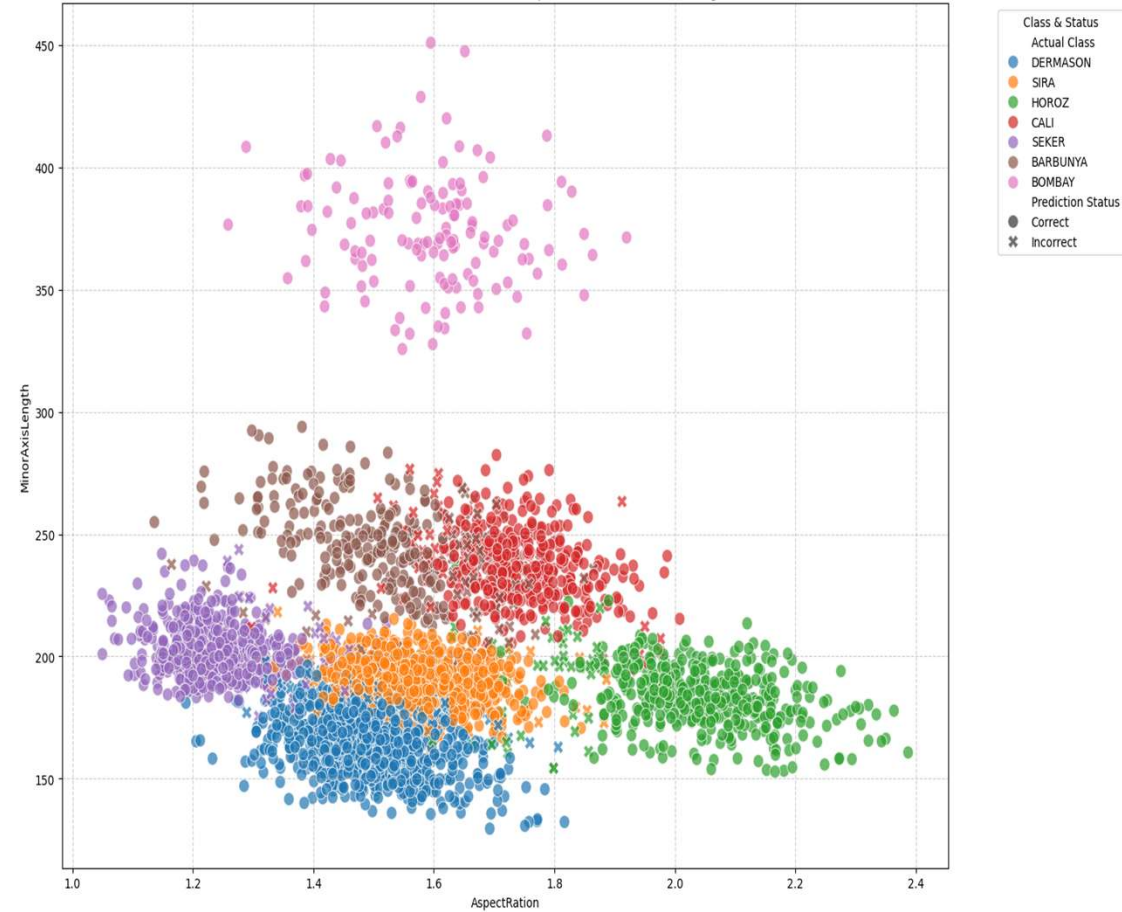
Rank	MRMR		ReliefF	
	Feature	MRMR score	Feature	ReliefF score
1	MajorAxisLength	0.9512	ShapeFactor1	0.1057
2	Extent	0.8748	ShapeFactor2	0.0904
3	Solidity	0.8334	Perimeter	0.0885
4	ShapeFactor1	0.8282	EquivDiameter	0.0884
5	AspectRatio	0.8065	MinorAxisLength	0.0874
6	ShapeFactor4	0.8065	MajorAxisLength	0.0865
7	roundness	0.7869	Area	0.0698
8	ShapeFactor2	0.7532	ConvexArea	0.0685
9	EquivDiameter	0.7173	Compactness	0.0607
10	MinorAxisLength	0.7173	ShapeFactor3	0.0598
11	Perimeter	0.6455	AspectRatio	0.0588
12	ShapeFactor3	0.6286	Extent	0.0544
13	ConvexArea	0.6286	roundness	0.0471
14	Compactness	0.5782	Eccentricity	0.0455
15	Area	0.5744	ShapeFactor4	0.0258
16	Eccentricity	0.5495	Solidity	0.0198

Classification using DT

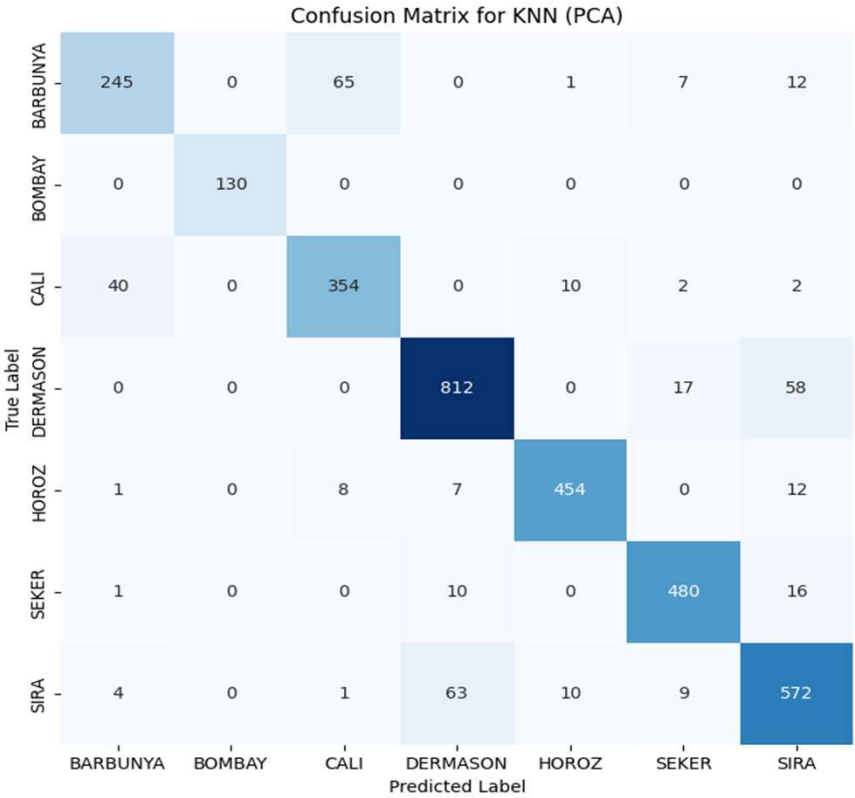
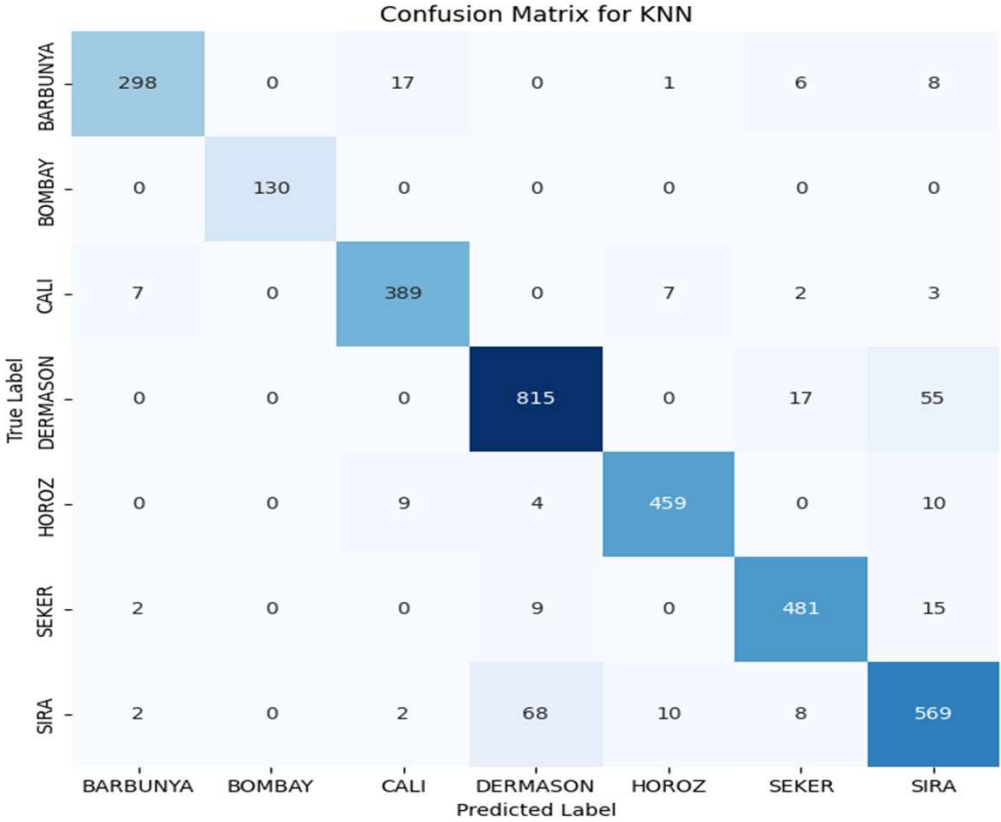
Confusion Matrix for Decision Tree



Predictions for Decision Tree (Features: AspectRatio vs MinorAxisLength)



Classification using DT

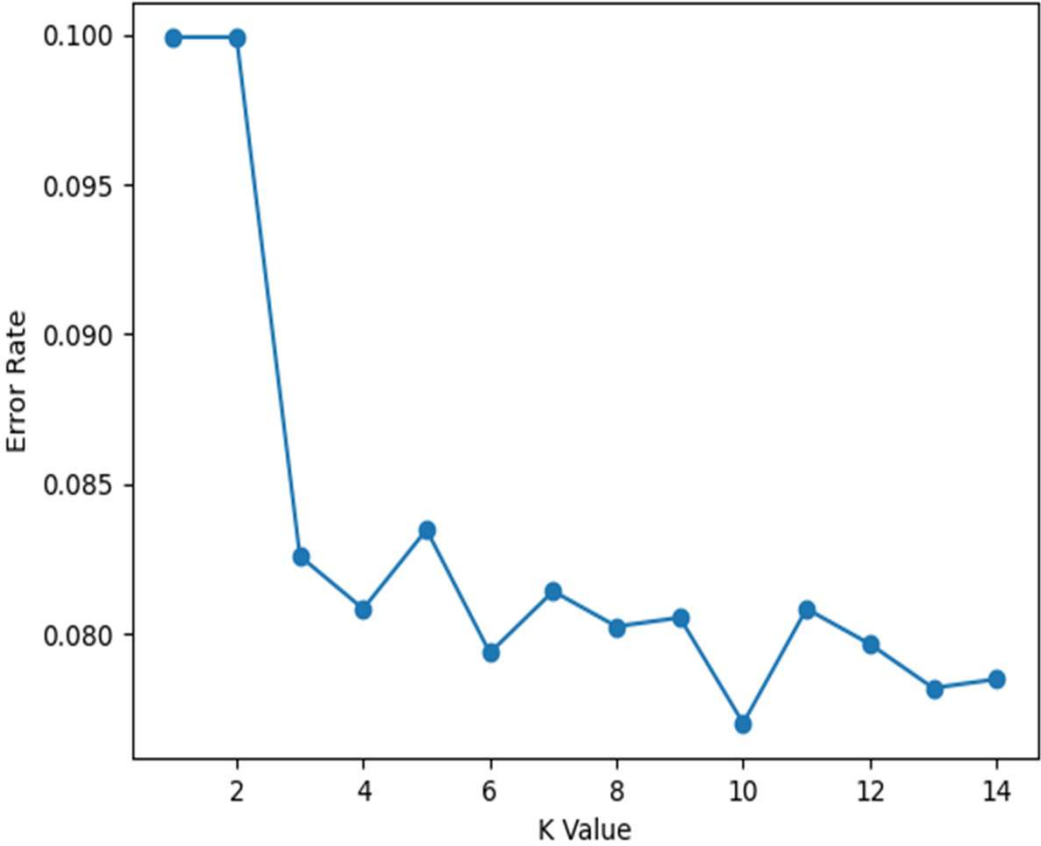


Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
88.481	0.889	0.885	0.885	0.978

Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
85.01	0.852	0.850	0.850	0.973

Classification using KNN

Elbow Method for Optimal K



Confusion Matrix for KNN

True Label	BARBUNYA	BOMBAY	CALI	DERMASON	HOROZ	SEKER	SIRA
BARBUNYA	298	0	17	0	1	6	8
BOMBAY	0	130	0	0	0	0	0
CALI	7	0	389	0	7	2	3
DERMASON	0	0	0	815	0	17	55
HOROZ	0	0	9	4	459	0	10
SEKER	2	0	0	9	0	481	15
SIRA	2	0	2	68	10	8	569
		Predicted Label					

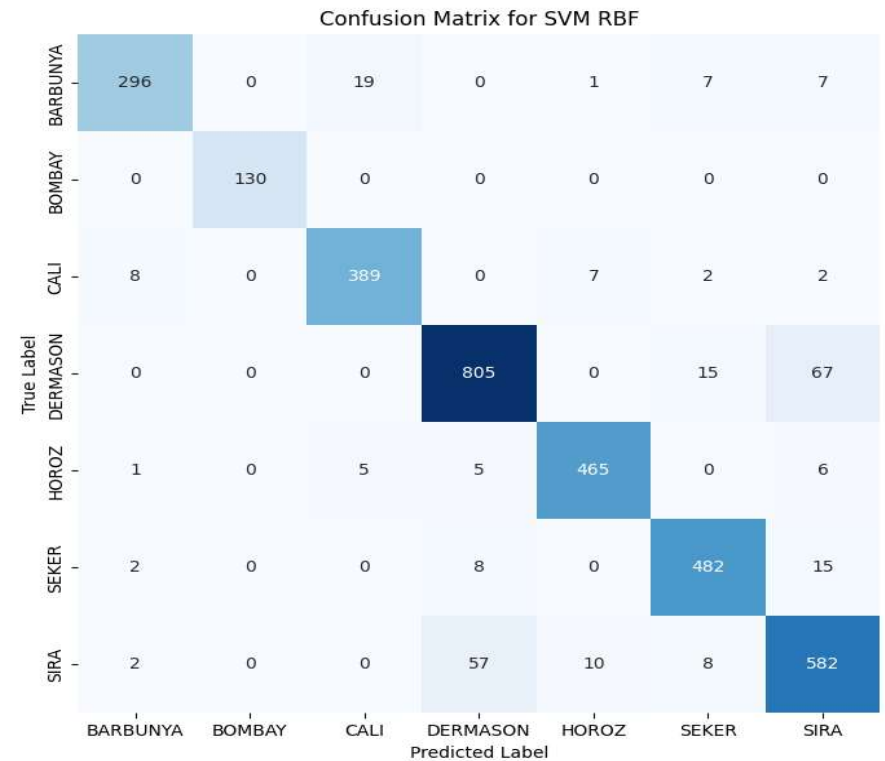
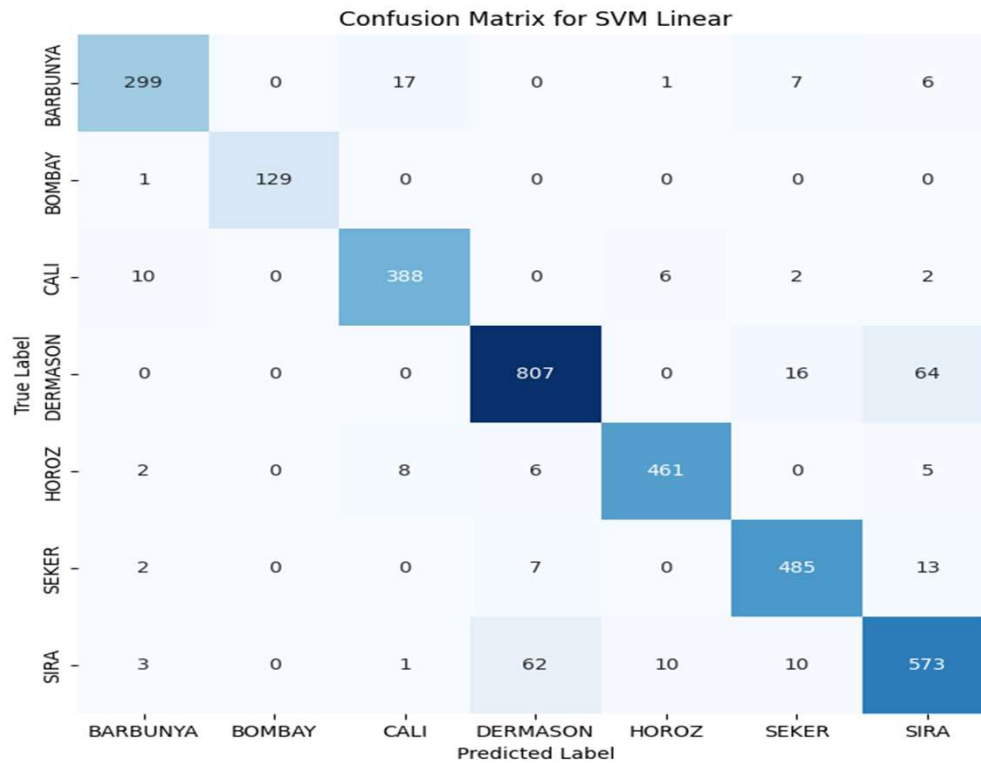
Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
92.30	0.923	0.923	0.923	0.988

Confusion Matrix for KNN (PCA)

True Label	BARBUNYA	BOMBAY	CALI	DERMASON	HOROZ	SEKER	SIRA
BARBUNYA	245	0	65	0	1	7	12
BOMBAY	0	130	0	0	0	0	0
CALI	40	0	354	0	10	2	2
DERMASON	0	0	0	812	0	17	58
HOROZ	1	0	8	7	454	0	12
SEKER	1	0	0	10	0	480	16
SIRA	4	0	1	63	10	9	572
		Predicted Label					

Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
89.53	0.895	0.895	0.894	0.982

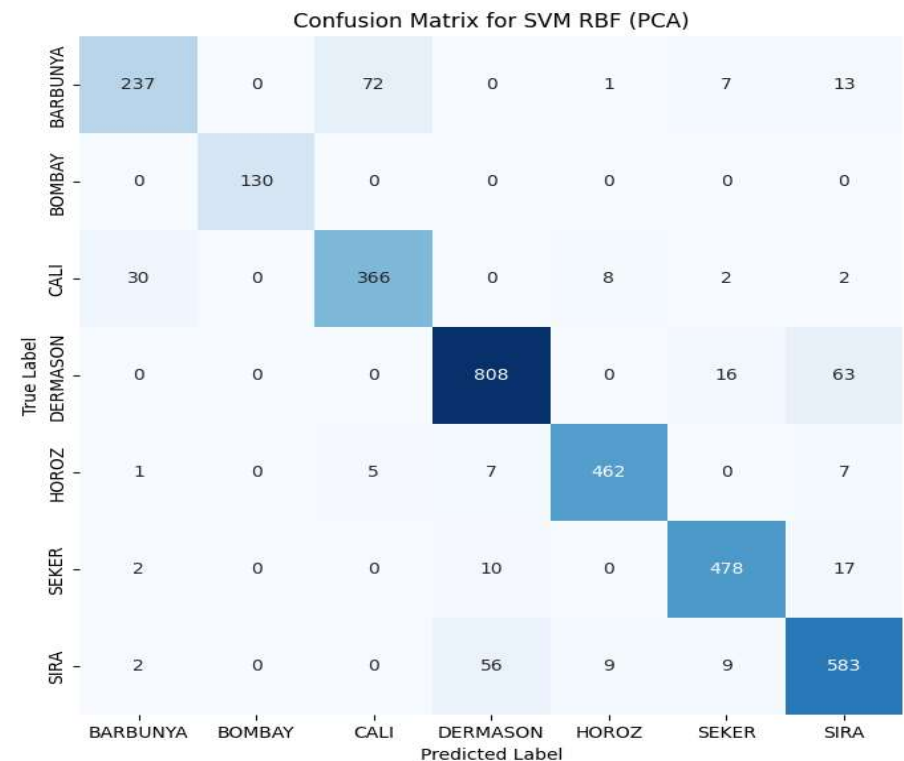
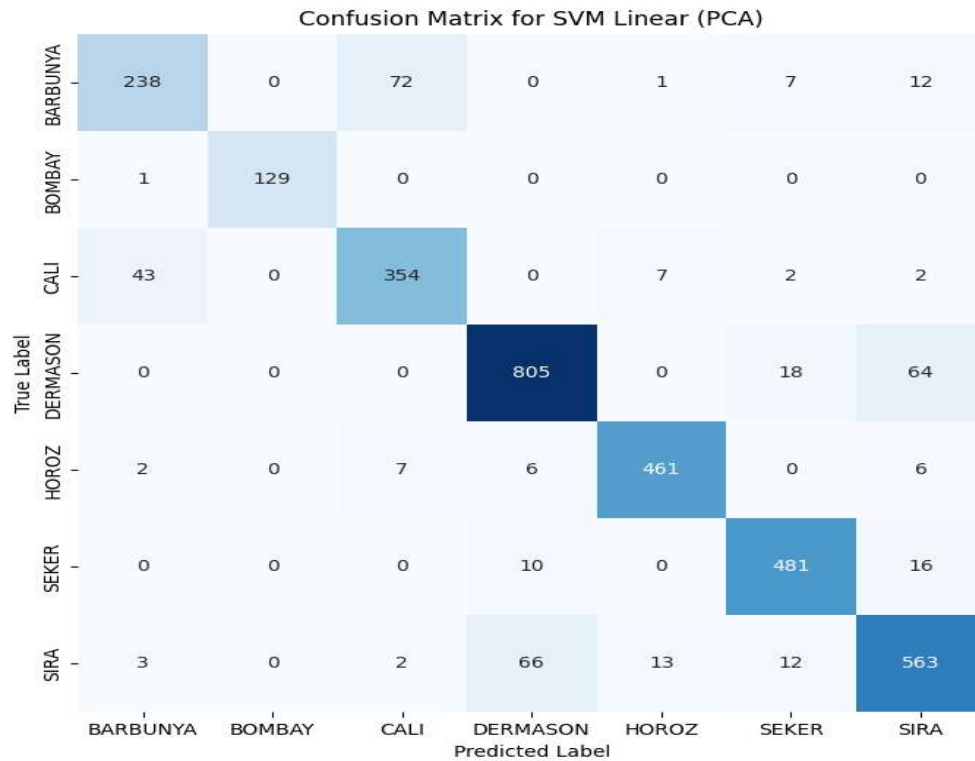
Classification using SVM



Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
92.33	0.923	0.922	0.923	0.995

Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
92.54	0.926	0.925	0.926	0.994

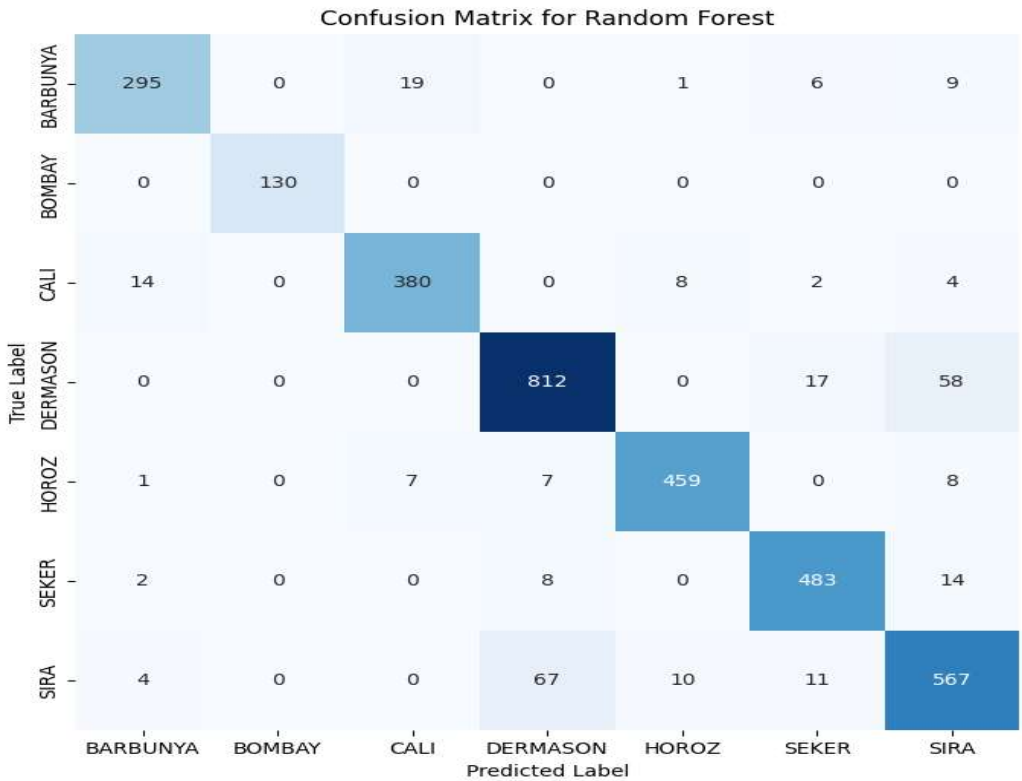
Classification using SVM



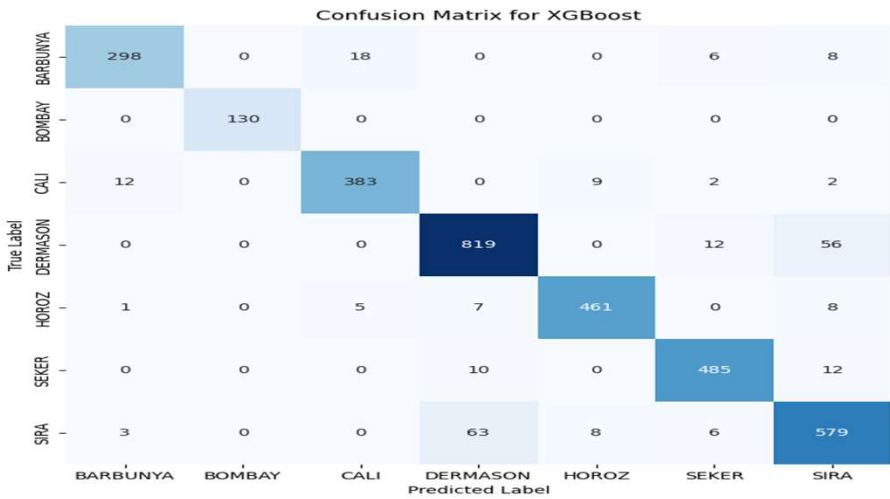
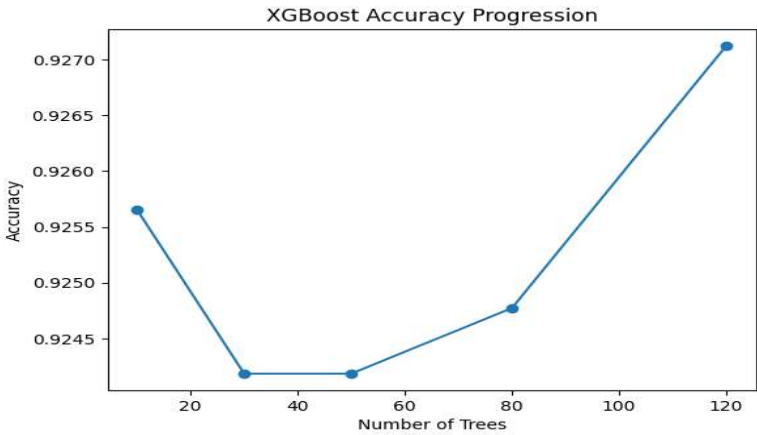
Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
89.06	0.890	0.891	0.890	0.990

Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
90.04	0.901	0.900	0.899	0.9912

Classification using Ensemble Methods



Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
91.86	0.919	0.918	0.918	0.9913



Accuracy (%)	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
92.71	0.9274	0.9271	0.9271	0.9950

Performance Comparison

Without PCA

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
DT	0.884808	0.888875	0.884808	0.884758	0.978576
KNN	0.923009	0.923323	0.923009	0.923038	0.988480
SVM Linear	0.923303	0.923447	0.923303	0.923304	0.995244
SVM RBF	0.925360	0.925882	0.925360	0.925464	0.994968
Random Forest	0.918601	0.918718	0.918601	0.918592	0.991329
XGBoost	0.927123	0.927410	0.927123	0.927192	0.995022

With PCA

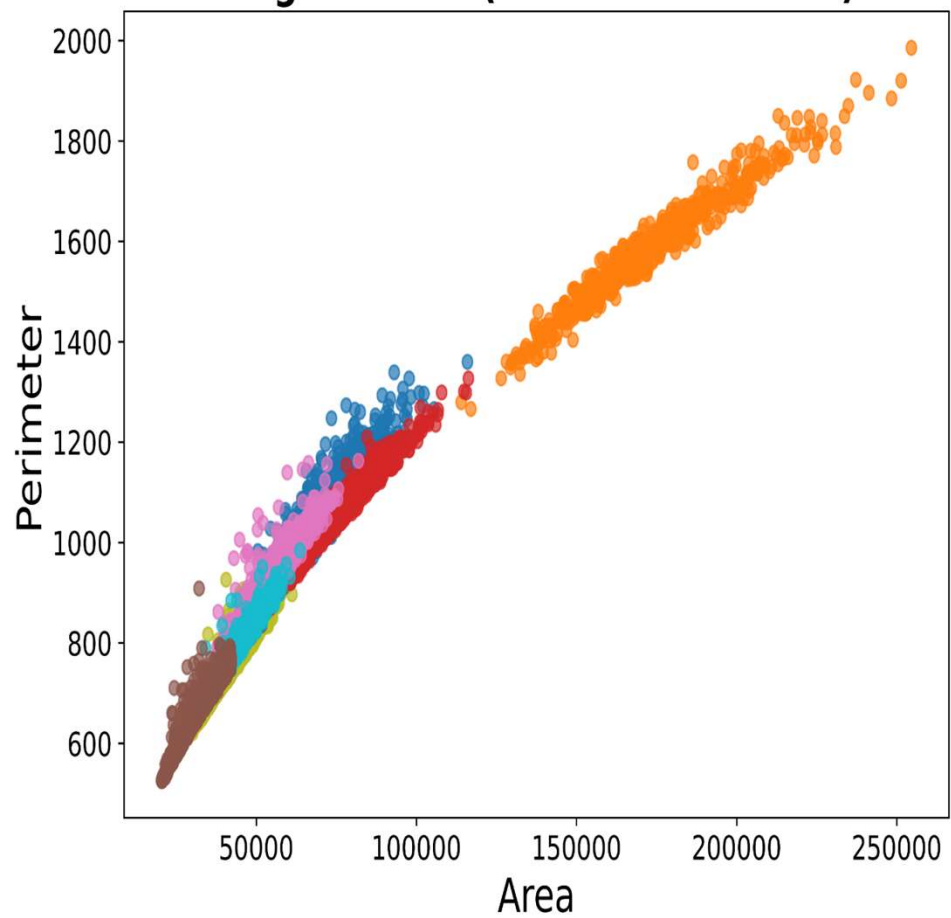
Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score	AUC
DT	0.850132	0.852259	0.850132	0.850848	0.973365
KNN	0.895386	0.895360	0.895386	0.894979	0.982626
SVM Linear	0.890685	0.890469	0.890685	0.890084	0.990942
SVM RBF	0.900382	0.901044	0.900382	0.899753	0.991248
Random Forest	0.897150	0.896822	0.897150	0.896731	0.988445
XGBoost	0.887158	0.886938	0.887158	0.886922	0.989264

Observations:

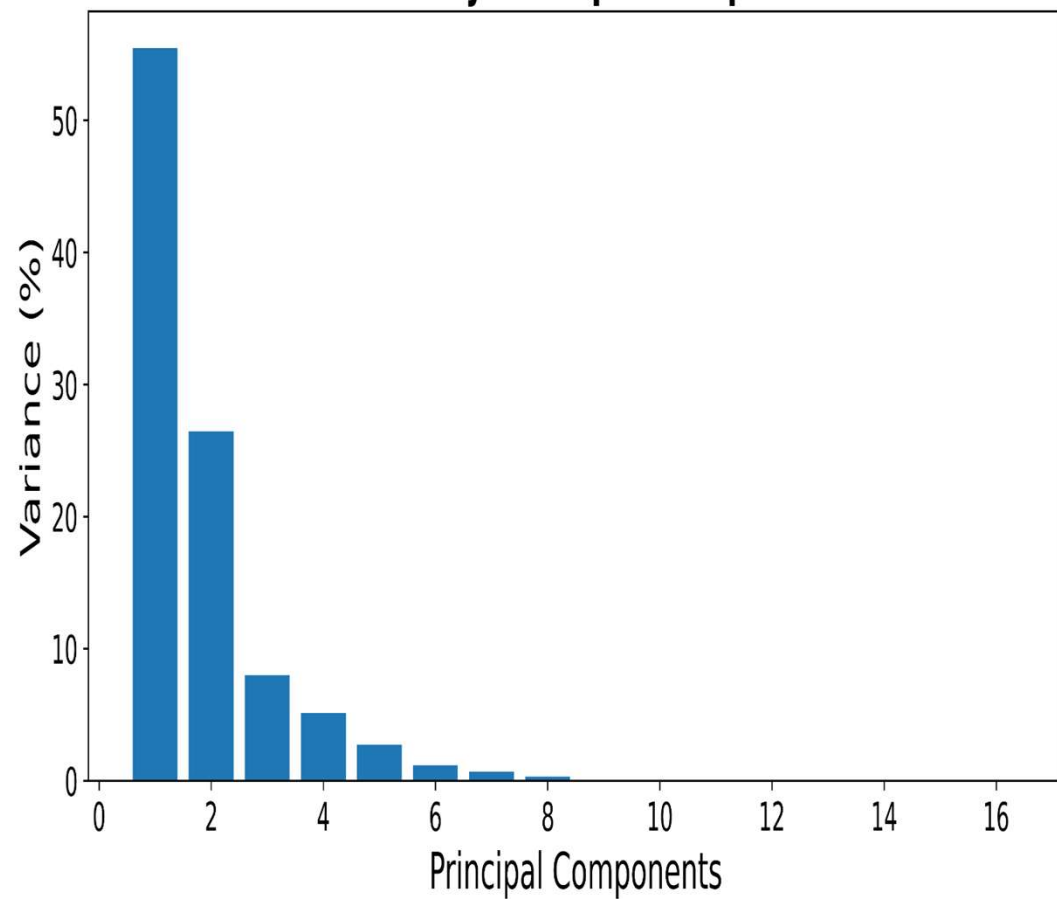
- ✓ PCA Achieved Significant Feature Reduction (75%) with Moderate Accuracy Loss (~2-3%)
- ✓ Original Feature Space Provides Superior Classification Performance (XGBoost (92.71%))
- ✓ PCA Offers a Favorable Accuracy–Complexity Tradeoff (SVM-RBF - 90.04%)

Clustering Algorithms

Original Data (Area vs Perimeter)

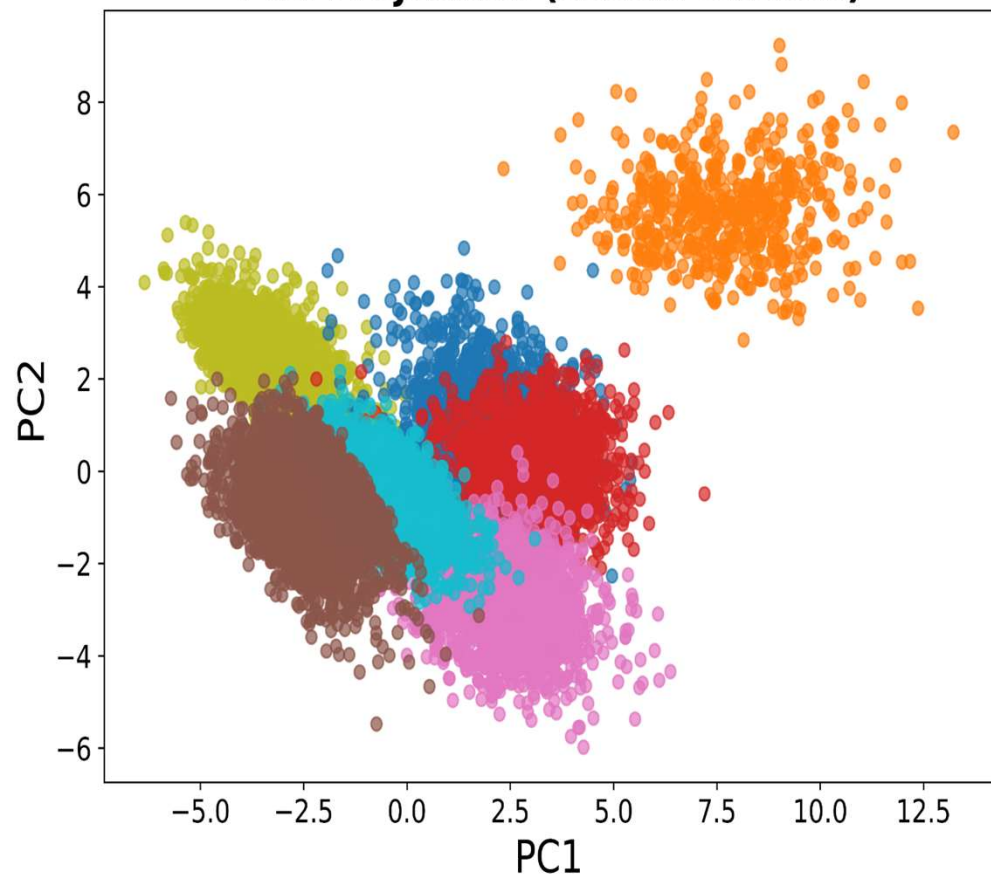


Variance by Principal Components

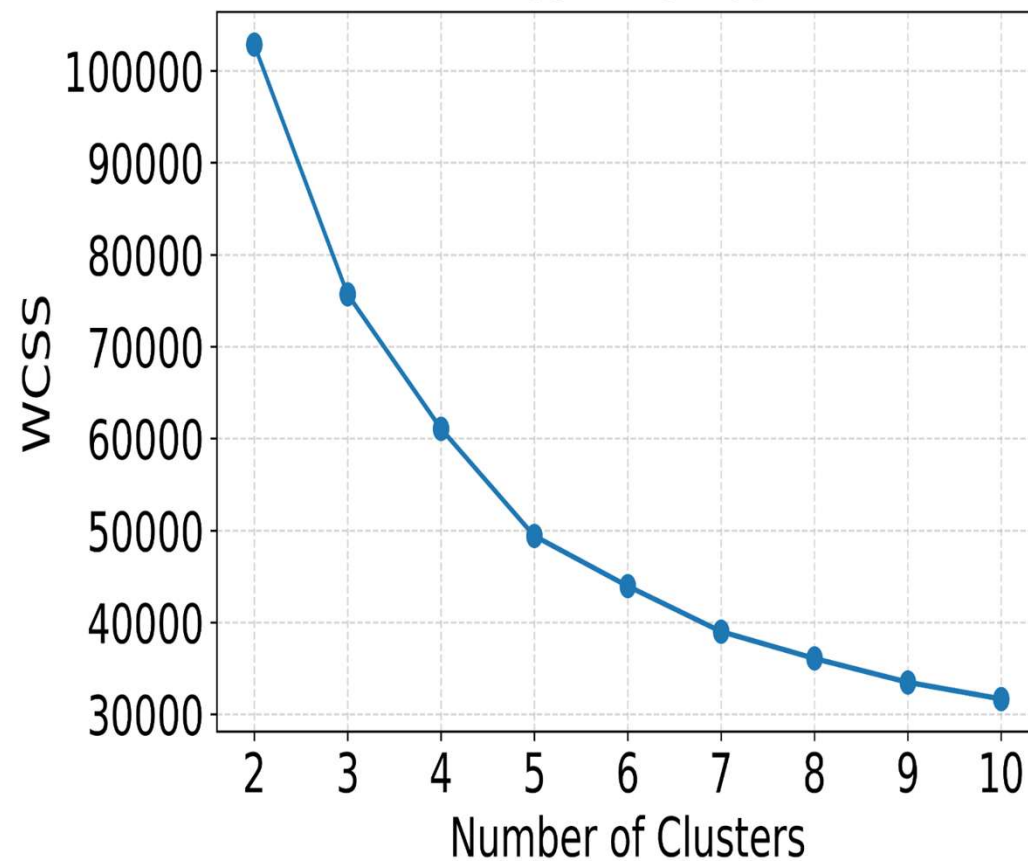


PCA and Elbow Method:

PCA Projection (Actual Classes)

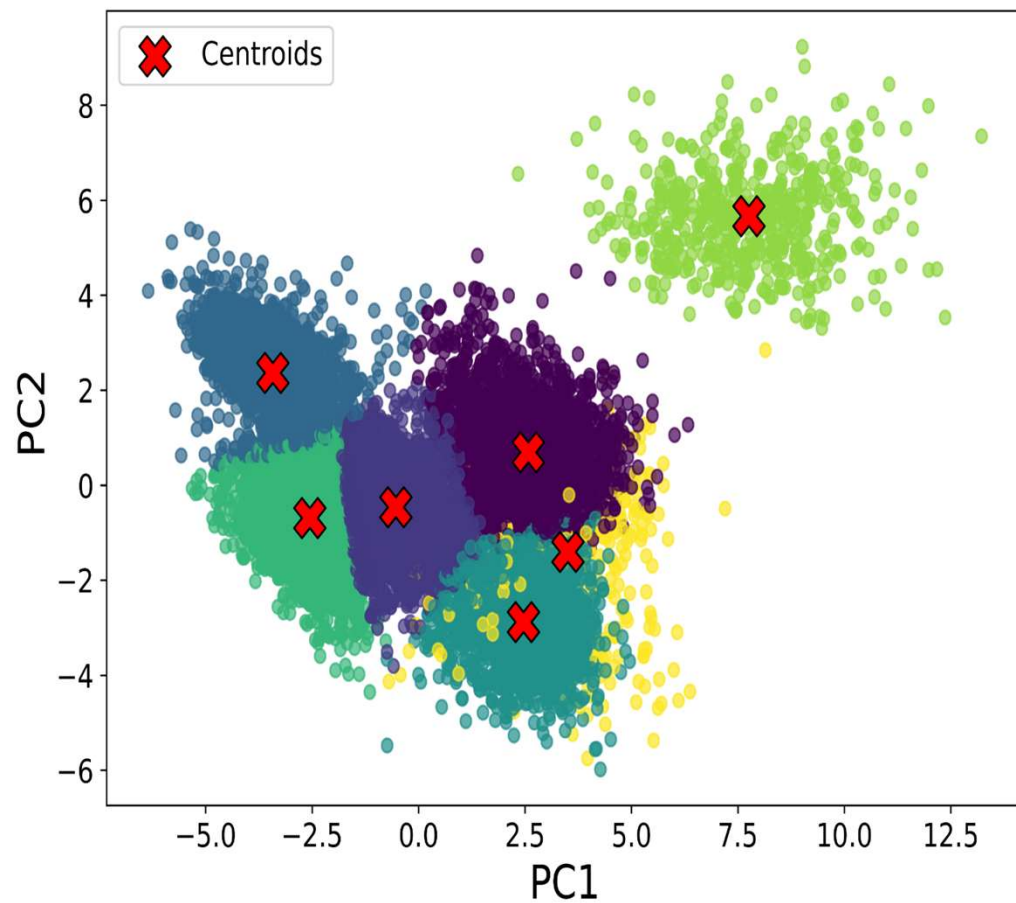


Elbow Method

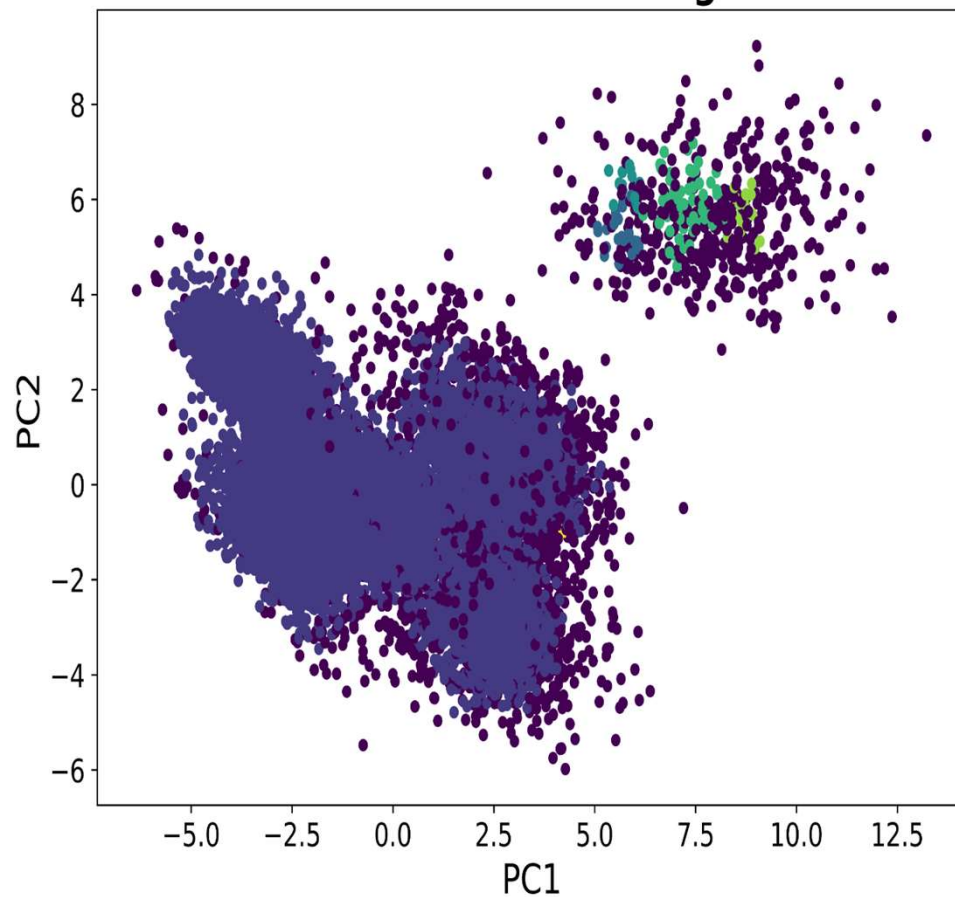


Clustering Methods:

KMeans Clusters

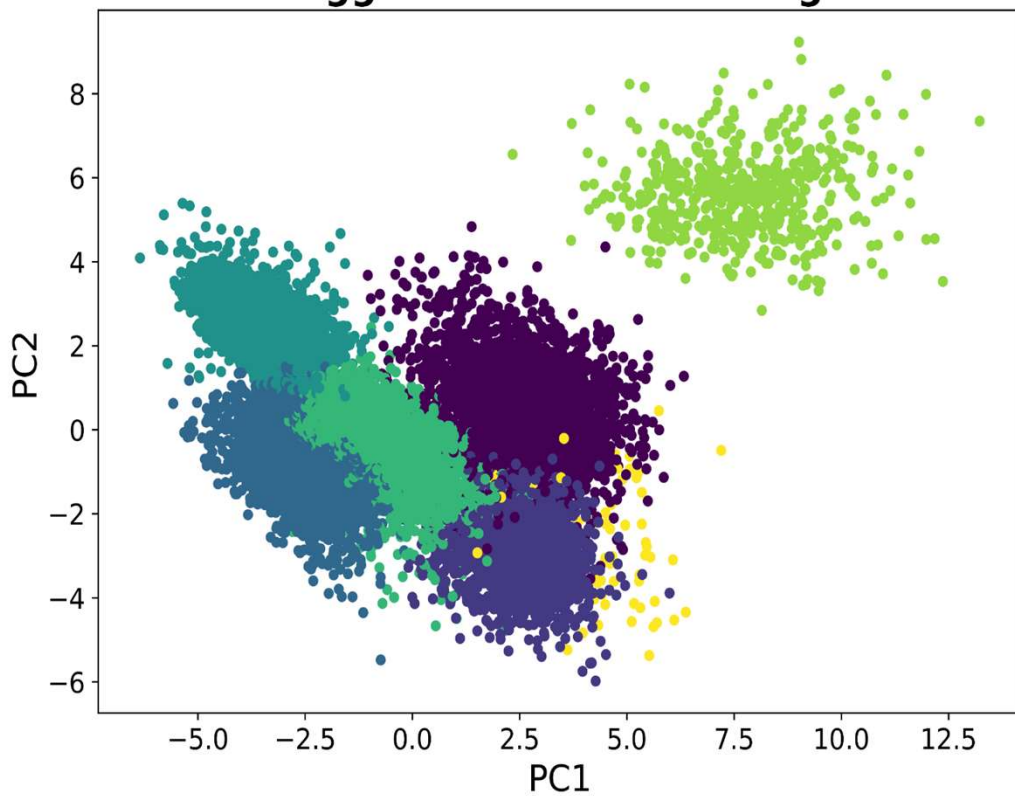


DBSCAN Clustering

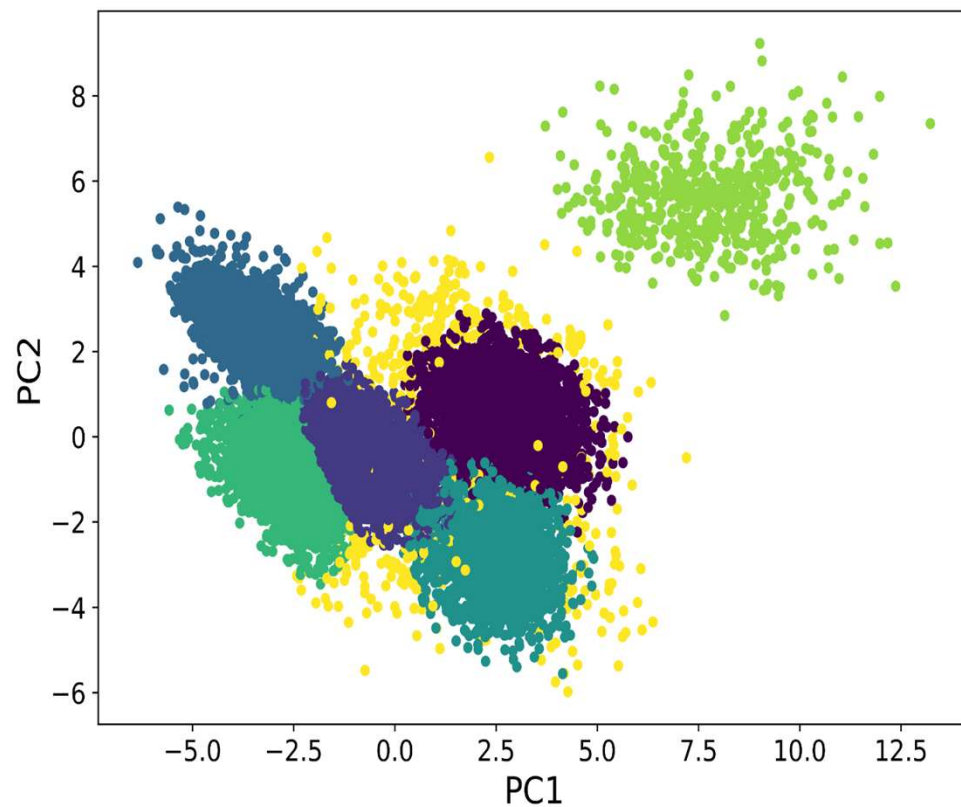


Clustering Methods:

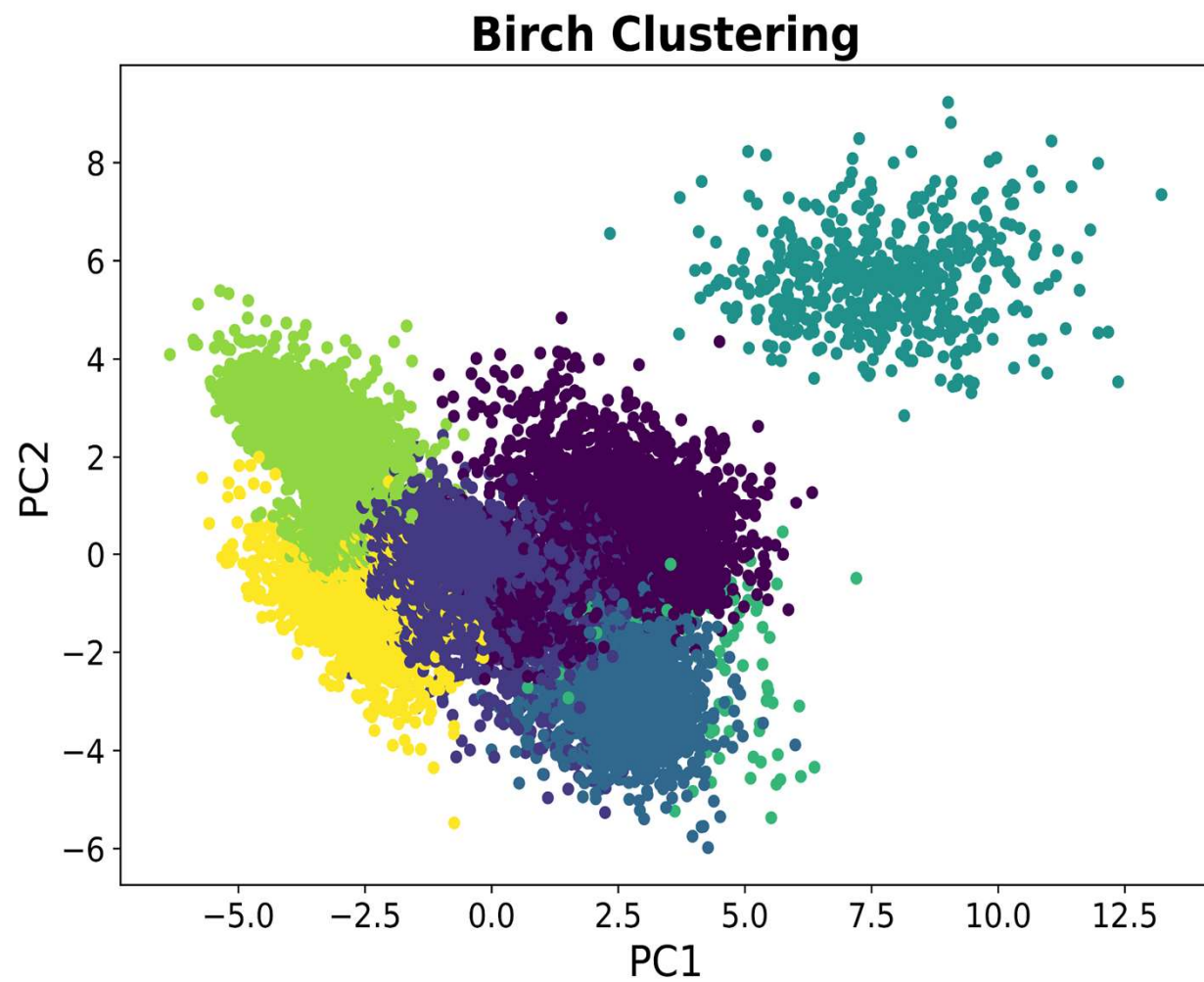
Agglomerative Clustering



Gaussian Mixture Model



Clustering Method:



Clustering Evaluation Metrics:

- Silhouette Score

- Range: -1 to +1
- Higher is better
- Measures cluster cohesion and separation

$$s = \frac{b - a}{\max(a, b)}$$

- Calinski–Harabasz Index

- Range: 0 to ∞
- Higher is better
- Ratio of between-cluster to within-cluster variance

$$CH = \frac{\text{Between-cluster variance}/(k - 1)}{\text{Within-cluster variance}/(n - k)}$$

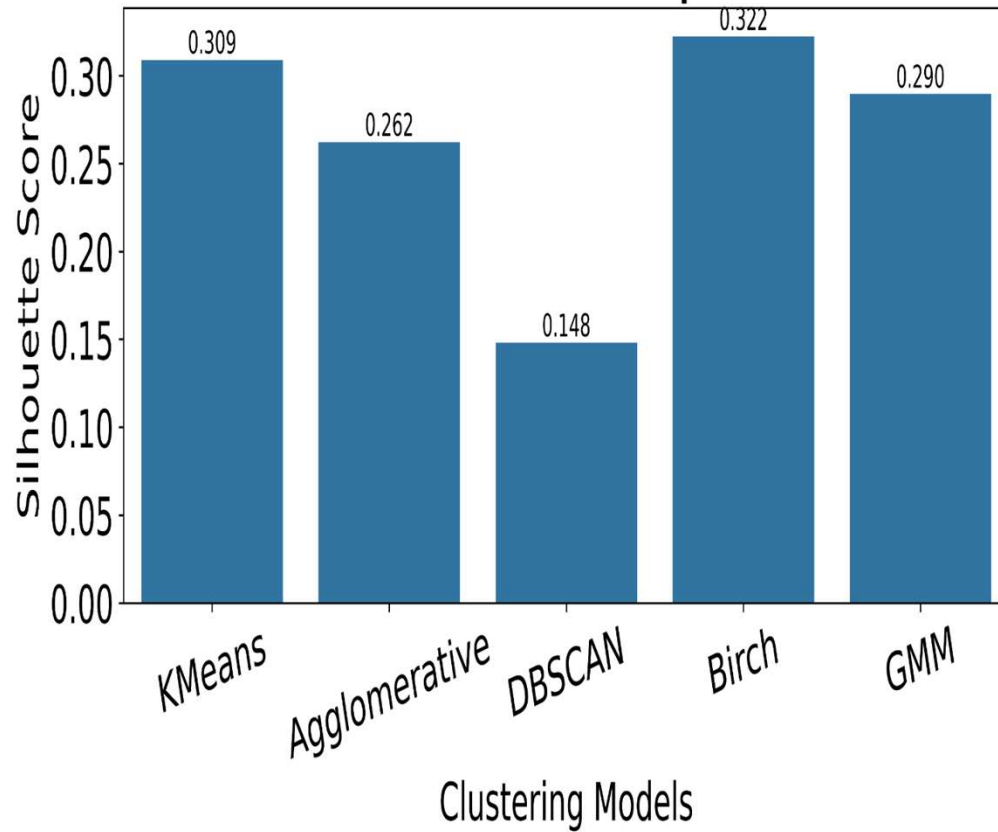
- Davies–Bouldin Index

- Range: 0 to ∞
- Lower is better
- Measures cluster similarity and overlap

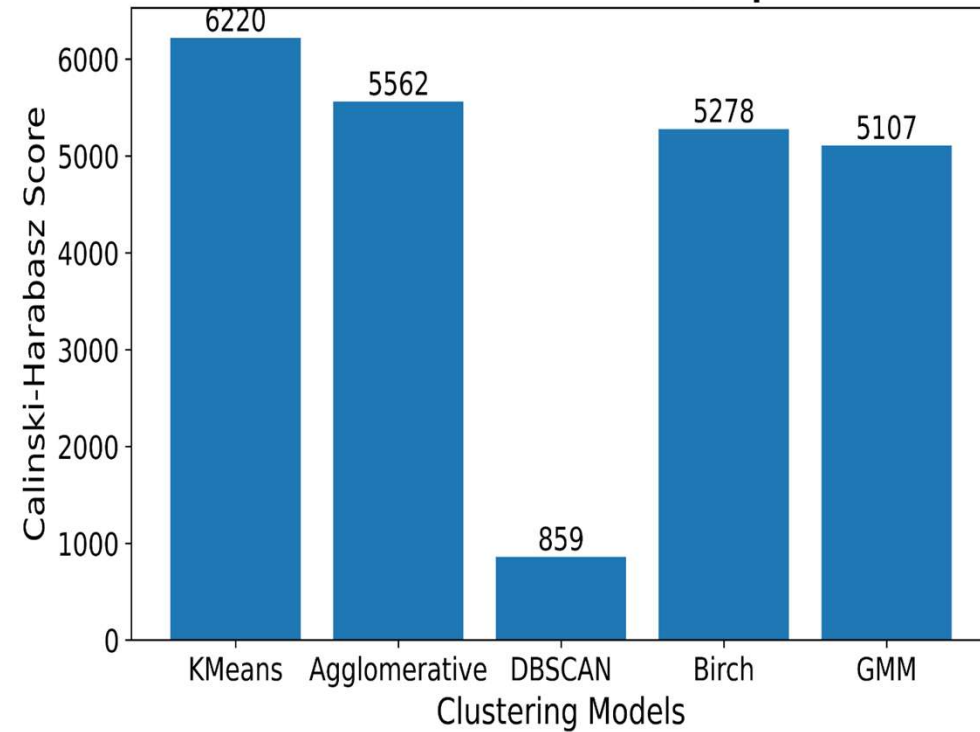
$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{S_i + S_j}{M_{ij}} \right)$$

Comparative Evaluation

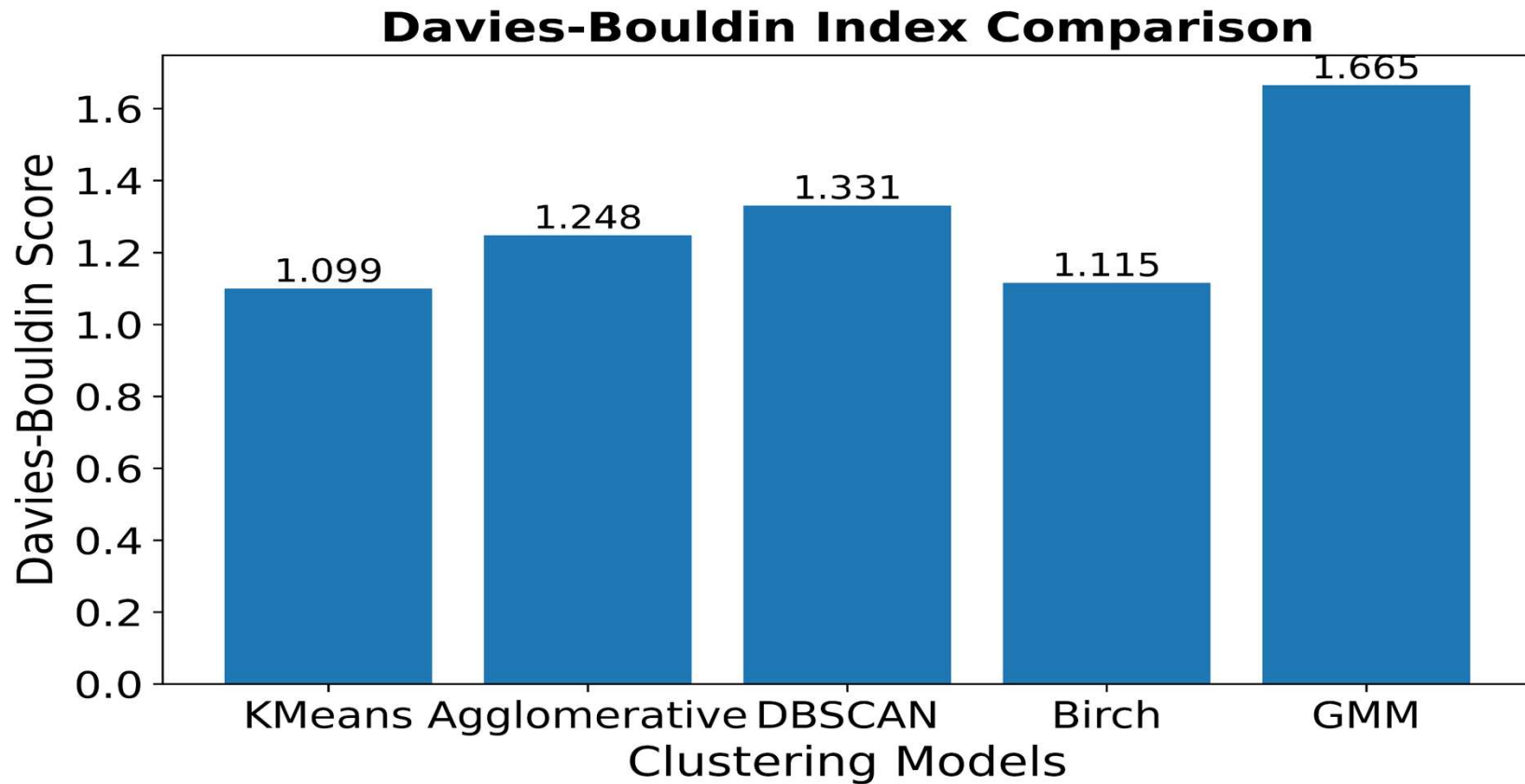
Silhouette Score Comparison



Calinski-Harabasz Index Comparison



Comparative Evaluation



Comparative Evaluation

Model	Silhouette Score ↑	Calinski–Harabasz Index ↑	Davies–Bouldin Index ↓
K-Means	0.309	6219.92	1.0994
Agglomerative	0.2622	5562.25	1.2477
DBSCAN	0.1481	858.78	1.3306
Birch	0.3224	5277.53	1.1150
GMM	0.2897	5107.09	1.6652

Observations:

- ✓ **PCA effectively reduced dimensionality**, preserving the most significant variance and enabling clear visualization of the bean dataset.
- ✓ **K-Means and Birch achieved the best clustering performance**, with K-Means obtaining the highest Calinski–Harabasz Index (6219.92) and lowest Davies–Bouldin Index (1.0994), while Birch achieved the highest Silhouette Score (0.3224).
- ✓ **The combination of PCA and clustering successfully identified natural groupings in the dataset**, demonstrating its effectiveness for unsupervised pattern and data analysis.

References

- ✓ <https://www.kaggle.com/datasets/muratkokludataset/dry-bean-dataset>
- ✓ KOKLU, M. and OZKAN, I.A., (2023), Multiclass Classification of Dry Beans Using Computer Vision and Machine Learning Techniques. Computers and Electronics in Agriculture, 174, 105507.
DOI: <https://doi.org/10.1016/j.compag.2020.105507>

Thank You